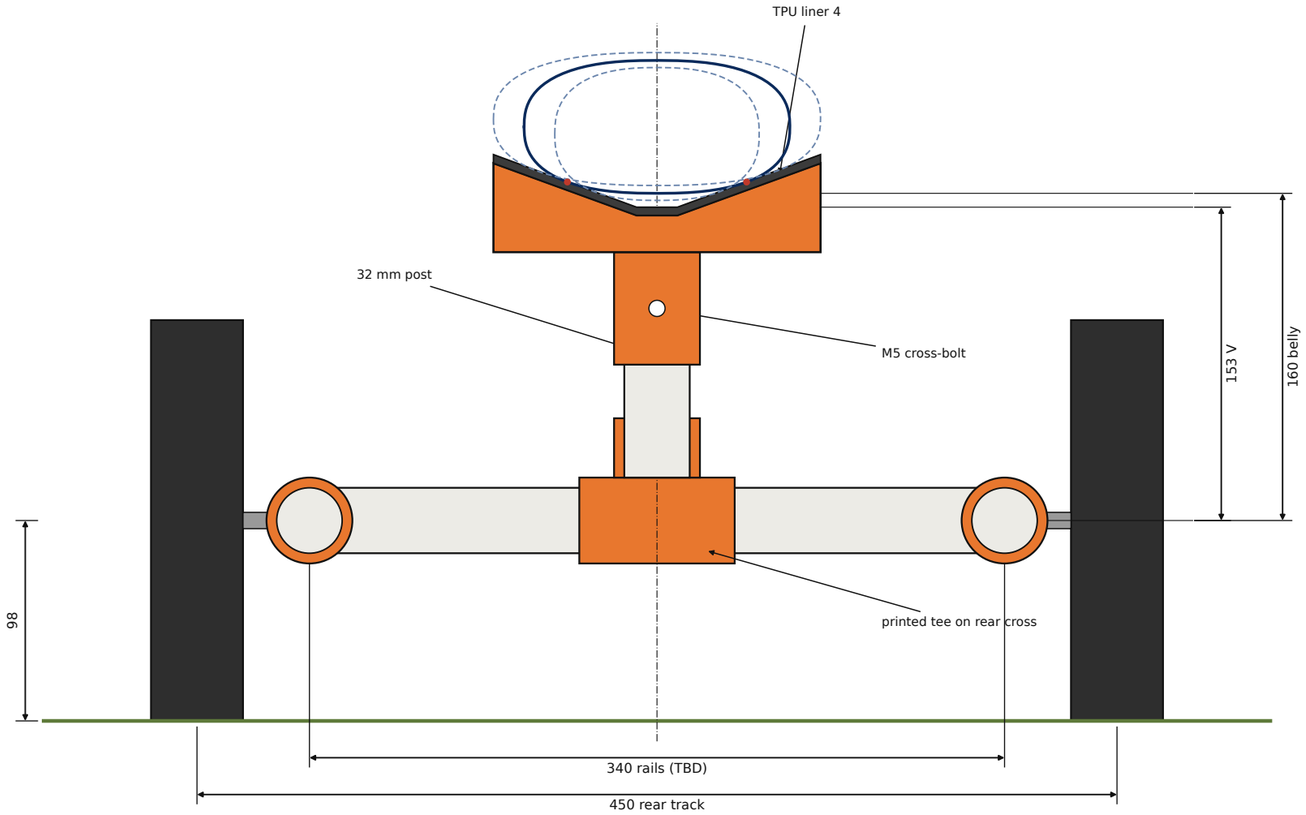
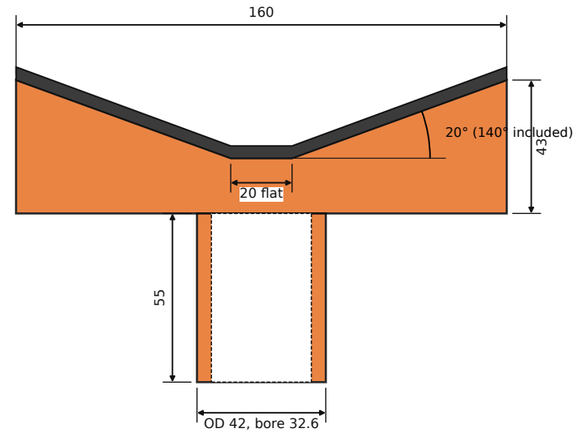
SECTION AT THE REAR AXLE, LOOKING FORWARD 1:3

fuselage section estimated (solid = 130 mm wide, dashed = 100 / 160 mm); red = contact lines



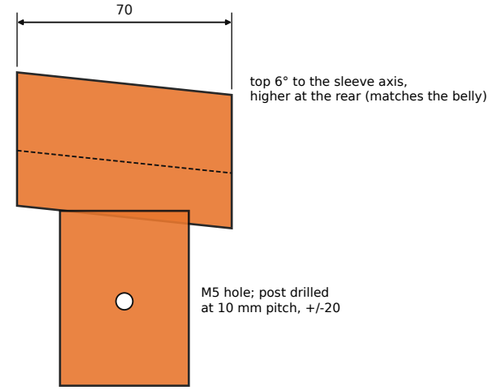
CRADLE SECTION 1:2

printed PETG body, TPU liner



CRADLE SIDE 1:2

nose to the right



KEY DATA	VALUE
V apex (TPU) above axle	153 mm (251 above ground)
Belly bottom above axle	160 mm (nominal section)
Cradle	160 wide x 70 long, 20 deg flanks, 20 flat
Post	32 mm tube, top 141 above axle; adjust +/-20 mm
Section assumed (TBD)	130 wide x 65 deep, rounded

NOTES

1. Cradle sits on a 32 mm post in a printed tee at the middle of the rear cross member, at the rear axle station (about 600 mm aft of the wing LE). The belly bottom sits 160 mm above the axle.
2. V-cradle, 20 deg flanks: self-centres the tail and opens upward, so the tail lifts straight out when it rises. Belly bottom sits 3-11 mm above the V apex for sections 100-160 mm wide (7 mm at 130).
3. Set height on the aircraft: wing seated on both trays, belly just bearing on the cradle without lifting the LEs. Measure the belly width and profile at the station and check pushrods/antennas clear the cradle (TBD).
4. Print PETG, 4 walls, 30% infill; liner TPU 95A (print or cut from 4 mm TPU sheet, bond with CA/contact cement).

Fuselage section at the station is estimated from the drawing (TBD) - measure before printing.

QUT AEROSPACE SOCIETY - BELIEVER UAV — **CONCEPT**

LAUNCH DOLLY	DWG No.
TAIL SUPPORT CRADLE	**BLV-DOL-007**

SCALE AS SHOWN (A3)	SHEET 1 OF 1	REV A 2026-11-05
UNITS mm | ANGLES deg	GENERATED: tail_cradle.py	PROJECTION